TEMCO LEARN

Register maps

TStat registers, part 3 · built Nov 13, 2026

TStat registers, part 3

The Modbus register map for the TStat, in address order. Addresses are decimal, and where the map gives a range the register occupies every address in it. This part begins at address 325.0 and ends at 444.0.

Address	Bytes	Register and description
325.0	1.0	Rotation time left. Long long time left the rotation will happen. READ ONLY.
326.0	1.0	Show the size of E2 chip. 0 = 24c02, 1 = 24c08/24c16.
327.0	1.0	Assign the timer to be used for which feature. 0 = period timer, 1 = rotation timer, 2 = interlock, 3 = PWM timer.
328.0	1.0	The output1 function, there are three functions for the output1.0 = normal ON/OFF output, 1 = rotation (old disabled feature) 2 = lighting control where one button can be assigned to toggle a relay on & off
329.0	1.0	Rotation Feature: Show which output table is used for output1 when output1 belongs to the rotation group
330.0	1.0	Rotation Feature: Show which output table is used for output1 when output2 belongs to the rotation group
331.0	1.0	Rotation Feature: Show which output table is used for output1 when output3 belongs to the rotation group
332.0	1.0	Rotation Feature: Show which output table is used for output1 when output4 belongs to the rotation group
333.0	2.0	How much time left before output rotation gets retriggered
334.0	1.0	The output2 function setting: 0 = normal ON/OFF output, 1 = rotation, 2 = lighting control.
335.0	1.0	Output3 function setting (see above)
336.0	1.0	Output4 function setting (see above)
337.0	1.0	Output5 function setting (see above)
338.0	1.0	Default occupied setpoint. Works in concert with the "occupied setpoint control register", register 339
339.0	1.0	Occupied Setpoint Control Register: 0 = normal, setpoint is managed by the serial port and keypad, the stat will remember the last occupied setpoint and use that during the next occupied period. 1 = Default mode, the last occupied setpoint if forgotten and the occupied setpoint gets reset to the default. 2 = trigger an event, when a master controller writes 2 to this register, the default setpoint will be copied to the occupied setpoint after which the Tstat will set the value back to 1 to show the event has been serviced.
340.0	1.0	Enable/disable PIR correspond 1/0 respectively.

Address	Bytes	Register and description
341.0	1.0	PWM output range in COAST mode. 0 = CLOSE, 1 = OPEN, 2 = 0-100%, 3 = 50-100%, 4 = 0-50%. MSb 4 bits correspond to output4 and LSB 4 bits correspond to output5
342.0	1.0	PWM output range in COOLING2 mode. 0 = CLOSE, 1 = OPEN, 2 = 0-100%, 3 = 50-100%, 4 = 0-50%. MSb 4 bits correspond to output4 and LSB 4 bits correspond to output5
343.0	1.0	PWM output range in COOLING3 mode. 0 = CLOSE, 1 = OPEN, 2 = 0-100%, 3 = 50-100%, 4 = 0-50%. MSb 4 bits correspond to output4 and LSB 4 bits correspond to output5
344.0	1.0	PWM output range in COOLING1 mode. 0 = CLOSE, 1 = OPEN, 2 = 0-100%, 3 = 50-100%, 4 = 0-50%. MSb 4 bits correspond to output4 and LSB 4 bits correspond to output5
345.0	1.0	PWM output range in HEATING1 mode. 0 = CLOSE, 1 = OPEN, 2 = 0-100%, 3 = 50-100%, 4 = 0-50%. MSb 4 bits correspond to output4 and LSB 4 bits correspond to output5
346.0	1.0	PWM output range in HEATING2 mode. 0 = CLOSE, 1 = OPEN, 2 = 0-100%, 3 = 50-100%, 4 = 0-50%. MSb 4 bits correspond to output4 and LSB 4 bits correspond to output5
347.0	1.0	PWM output range in HEATING3 mode. 0 = CLOSE, 1 = OPEN, 2 = 0-100%, 3 = 50-100%, 4 = 0-50%. MSb 4 bits correspond to output4 and LSB 4 bits correspond to output5
348.0	1.0	The ON period take how many percentage for output4
349.0	1.0	The ON period take how many percentage for output5
350.0	1.0	Free cooling configuration. bit0, free cool enable/disable, 0 = disable, 1= enable. bit1, free cool available decided by local or external master. 0 = local, 1 = NC. bit2, free cool available status, 0 = npt available, 1= available. bit4, show the status if NC is OK when the free cool decided by NC.
351.0	1.0	Analog output OFF table, coasting mode
352.0	1.0	Analog output OFF table, cooling1 mode
353.0	1.0	Analog output OFF table, cooling2 mode
354.0	1.0	Analog output OFF table, cooling3 mode
355.0	1.0	Analog output OFF table, heating1 mode
356.0	1.0	Analog output OFF table, heating2 mode

Address	Bytes	Register and description
357.0	1.0	Analog output OFF table, heating3 mode — The following register for Tstat 5E
358.0	1.0	Register lock. All registers except fan speed and manual inputs/outputs register are not writable. 0 = lock,1 = no lock. — Bytes — Description
359.0	1.0	Outside temperature for free cooling,from external sensor or NC. — 1.0 — ANALOG INPUT RANGE. 0 = raw data, 1 = thermistor, 2 = %, 3 = ON/OFF, 4 = CUSTOMER, 5 = OFF/ON
360.0	2.0	If outside temp be set from NC. The communication with NC must be set in this time,otherwise will set error status and use external sensor. — 1.0 — ANALOG INPUT RANGE. 0 = raw data, 1 = thermistor, 2 = %, 3 = ON/OFF, 4 = CUSTOMER, 5 = OFF/ON
361.0	1.0	If the outside air temp is lower than the room temperature by this amount, then the free cooling is worthwhile, 350 bit2 = 1.If the OAT is greater than the room temp, then free cooling mode is not worthwhile. , 350 bit2 = 0 — 1.0 — ANALOG INPUT RANGE. 0 = raw data, 1 = thermistor, 2 = %, 3 = ON/OFF, 4 = N/A, 5 = OFF/ON
362.0	1.0	Free cooling output configuration. Coasting mode — 1.0 — ANALOG INPUT RANGE. 0 = raw data, 1 = thermistor, 2 = %, 3 = ON/OFF, 4 = N/A, 5 = OFF/ON
363.0	1.0	Free cooling output configuration.Cooling1 mode — 1.0 — ANALOG INPUT RANGE. 0 = raw data, 1 = thermistor, 2 = %, 3 = ON/OFF, 4 = N/A, 5 = OFF/ON
364.0	1.0	Free cooling output configuration.Cooling2 mode — 1.0 — ANALOG INPUT RANGE. 0 = raw data, 1 = thermistor, 2 = %, 3 = ON/OFF, 4 = N/A, 5 = OFF/ON
365.0	1.0	Free cooling output configuration.Cooling3 mode — 1.0 — ANALOG INPUT RANGE. 0 = raw data, 1 = thermistor, 2 = %, 3 = ON/OFF, 4 = N/A, 5 = OFF/ON
366.0	1.0	Free cooling output configuration.Heating1 mode — 1.0 — ANALOG INPUT RANGE. 0 = raw data, 1 = thermistor, 2 = %, 3 = ON/OFF, 4 = N/A, 5 = OFF/ON
367.0	1.0	Free cooling output configuration.Heating2 mode — 2.0 — Analog input1 value
368.0	1.0	Free cooling output configuration.heating3 mode — 2.0 — Analog input2 value
369.0	1.0	Min Air,the units is percent. Set the minimum output for free cooling, the default is 15% — 2.0 — Analog input3 value
370.0	1.0	Outside air temperature in hottest day — 2.0 — Analog input4 value
371.0	1.0	Outside air temperature in coldest day — 2.0 — Analog input5 value
372.0	1.0	Offset in hottest day — 2.0 — Analog input6 value
373.0	1.0	Offset in coldest day — 2.0 — Analog input7 value

Address	Bytes	Register and description
374.0	2.0	Store setpoint in two bytes, the resolution is 0.1 — 2.0 — Analog input8 value
375.0	1.0	Crrent setpoint = user setpoint + offset setpoint — 2.0 — Calibration for analog input1
376.0	1.0	Setpoint offset — 2.0 — Calibration for analog input2
377.0	1.0	Change over sensor mode, 1 = cooling mode, 0 = heating mode. — 2.0 — Calibration for analog input3
378.0	1.0	external sensor inuse,0 = not use,1 = in use — 2.0 — Calibration for analog input4
379.0	1.0	external sensor status,0 = OK,1 = not OK,damaged or short circuit — 2.0 — Calibration for analog input5
380.0	2.0	day cooling setpoint,work together with heating setpoint in day office mode — 2.0 — Calibration for analog input6
381.0	1.0	daysetpoint option,0 =office,1 = hotel — 2.0 — Calibration for analog input7
382.0	1.0	Timer left for output1 ON to OFF delay — 2.0 — Calibration for analog input8
383.0	1.0	Timer left for output2 ON to OFF delay — 1.0 — The first character on LCD line 1, ASCII code
384.0	1.0	Timer left for output3 ON to OFF delay — 1.0 — The second character on LCD line 1, ASCII code
385.0	1.0	Timer left for output4 ON to OFF delay — 1.0 — The third character on LCD line 1, ASCII code
386.0	1.0	Timer left for output5 ON to OFF delay — 1.0 — The fourth character on LCD line 1, ASCII code
387.0	1.0	Timer left for output1 OFF to ON delay — 1.0 — The fifth character on LCD line 1, ASCII code
388.0	1.0	Timer left for output2 OFF to ON delay — 1.0 — The sixth character on LCD line 1, ASCII code
389.0	1.0	Timer left for output3 OFF to ON delay — 1.0 — The seventh character on LCD line 1, ASCII code
390.0	1.0	Timer left for output4 OFF to ON delay — 1.0 — The eighth character on LCD line 1, ASCII code
391.0	1.0	Timer left for output5 OFF to ON delay — 1.0 — The first character on LCD line 2, ASCII code

Address	Bytes	Register and description
392.0	1.0	The second character on LCD line 2, ASCII code
393.0	1.0	The third character on LCD line 2, ASCII code
394.0	1.0	The fourth character on LCD line 2, ASCII code
395.0	1.0	The fifth character on LCD line 2, ASCII code
396.0	1.0	The sixth character on LCD line 2, ASCII code
397.0	1.0	The seventh character on LCD line 2, ASCII code
398.0	1.0	The eighth character on LCD line 2, ASCII code
399.0	1.0	LCD turn off delay, 0 - 255 minutes
400.0	1.0	Select what parameter will display, 0 = nothing, 1 =temperature, 2 = setpoint, 3 through 10 correspond to input 1 through 8, 11 = mode
401.0	1.0	Select what parameter will display, 0 = nothing, 1 =temperature, 2 = setpoint, 3 through 10 correspond to input 1 through 8, 11 = mode
402.0	1.0	DAY icon control register. 0 = AUTO, 1 = turn OFF, 2 = turn ON.
403.0	1.0	NIGHT icon control register. 0 = AUTO, 1 = turn OFF, 2 = turn ON.
404.0	1.0	Current mode,0 = coasting,1 = heating,2 = cooling
405.0	1.0	Output 1 from ON to OFF time left
406.0	1.0	Output 2 from ON to OFF time left
407.0	1.0	Output 3 from ON to OFF time left
408.0	1.0	Output 4 from ON to OFF time left
409.0	1.0	Output 5 from ON to OFF tilm left
410.0	1.0	Output 1 from OFF to ON time left
411.0	1.0	Output 2 from OFF to ON time left
412.0	1.0	Output 3 from OFF to ON time left
413.0	1.0	Output 4 from OFF to ON time left
414.0	1.0	Output 5 from OFF to ON time left
415.0	1.0	The hottest output air temperature

Address	Bytes	Register and description
416.0	1.0	The coldest output air temperature
417.0	1.0	The offset for the hottest temperature
418.0	1.0	The offset for the coldest temperature
419.0	1.0	Crrent setpoint = user setpoint + offset setpoint
420.0	1.0	The current setpoint offset
421.0	1.0	turn off all fan icons
422.0	2.0	normal cooling setpoint(135) + cooling deadband
423.0	1.0	day setpointoption,1 = hotel,0 = office
424.0	1.0	Window interlock delay,the units is minute, from 0 to 255 minutes
425.0	2.0	User name. line 1:characters 1 & 2
426.0	2.0	User name. line 1:characters 3 & 4
427.0	2.0	User name. line 1:characters 5 & 6 — User name. line 1:characters 5 & 6
428.0	2.0	User name. line 1:characters 7 & 8 — User name. line 1:characters 7 & 8
429.0	2.0	User name. line 2:characters 1 & 2 — User name. line 2:characters 1 & 2
430.0	2.0	User name. line 2:characters 3 & 4 — User name. line 2:characters 3 & 4
431.0	2.0	User name. line 2:characters 5 & 6 — User name. line 2:characters 5 & 6
432.0	2.0	User name. line 2:characters 7 & 8 — User name. line 2:characters 7 & 8
433.0	2.0	Internal sensor name. line 1:characters 1 & 2 — Internal sensor name. line 1:characters 1 & 2 — Internal sensor name. line 1:characters 1 & 2 — Internal sensor name. line 1:characters 1 & 2 — Internal sensor name. line 1:characters 1 & 2 — Internal sensor name. line 1:characters 1 & 2
434.0	2.0	Internal sensor name. line 1:characters 3 & 4 — Internal sensor name. line 1:characters 3 & 4 — Internal sensor name. line 1:characters 3 & 4 — Internal sensor name. line 1:characters 3 & 4 — Internal sensor name. line 1:characters 3 & 4 — Internal sensor name. line 1:characters 3 & 4
435.0	2.0	Internal sensor name. line 1:characters 5 & 6 — Internal sensor name. line 1:characters 5 & 6 — Internal sensor name. line 1:characters 5 & 6 — Internal sensor name. line 1:characters 5 & 6 — Internal sensor name. line 1:characters 5 & 6 — Internal sensor name. line 1:characters 5 & 6

Address	Bytes	Register and description
436.0	2.0	Internal sensor name. line 1:characters 7 & 8 — Internal sensor name. line 1:characters 7 & 8 — Internal sensor name. line 1:characters 7 & 8 — Internal sensor name. line 1:characters 7 & 8 — Internal sensor name. line 1:characters 7 & 8 — Internal sensor name. line 1:characters 7 & 8
437.0	2.0	Input 1 user name. line 1:characters 1 & 2 — Input 1 user name. line 1:characters 1 & 2 — Input 1 user name. line 1:characters 1 & 2 — Input 1 user name. line 1:characters 1 & 2 — Input 1 user name. line 1:characters 1 & 2 — Input 1 user name. line 1:characters 1 & 2
438.0	2.0	Input 1 user name. line 1:characters 3 & 4 — Input 1 user name. line 1:characters 3 & 4 — Input 1 user name. line 1:characters 3 & 4 — Input 1 user name. line 1:characters 3 & 4 — Input 1 user name. line 1:characters 3 & 4 — Input 1 user name. line 1:characters 3 & 4
439.0	2.0	Input 1 user name. line 1:characters 5 & 6 — Input 1 user name. line 1:characters 5 & 6 — Input 1 user name. line 1:characters 5 & 6 — Input 1 user name. line 1:characters 5 & 6 — Input 1 user name. line 1:characters 5 & 6 — Input 1 user name. line 1:characters 5 & 6
440.0	2.0	Input 1 user name. line 1:characters 7 & 8 — Input 1 user name. line 1:characters 7 & 8 — Input 1 user name. line 1:characters 7 & 8 — Input 1 user name. line 1:characters 7 & 8 — Input 1 user name. line 1:characters 7 & 8 — Input 1 user name. line 1:characters 7 & 8
441.0	2.0	Input 2 user name. line 1:characters 1 & 2 — Input 2 user name. line 1:characters 1 & 2 — Input 2 user name. line 1:characters 1 & 2 — Input 2 user name. line 1:characters 1 & 2 — Input 2 user name. line 1:characters 1 & 2 — Input 2 user name. line 1:characters 1 & 2
442.0	2.0	Input 2 user name. line 1:characters 3 & 4 — Input 2 user name. line 1:characters 3 & 4 — Input 2 user name. line 1:characters 3 & 4 — Input 2 user name. line 1:characters 3 & 4 — Input 2 user name. line 1:characters 3 & 4 — Input 2 user name. line 1:characters 3 & 4
443.0	2.0	Input 2 user name. line 1:characters 5 & 6 — Input 2 user name. line 1:characters 5 & 6 — Input 2 user name. line 1:characters 5 & 6 — Input 2 user name. line 1:characters 5 & 6 — Input 2 user name. line 1:characters 5 & 6 — Input 2 user name. line 1:characters 5 & 6
444.0	2.0	Input 2 user name. line 1:characters 7 & 8 — Input 2 user name. line 1:characters 7 & 8 — Input 2 user name. line 1:characters 7 & 8 — Input 2 user name. line 1:characters 7 & 8 — Input 2 user name. line 1:characters 7 & 8 — Input 2 user name. line 1:characters 7 & 8